AliasGuard: separating alias contamination from passband distortion in sampled modal vibration

Computational Mechanics Projects

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Abstract

Downsampling can cause two structural vibration components to occupy the same discrete frequency. A filter may suppress this contamination while also biasing the desired mode. This reproducible study separates those effects for an exact three-mode oscillator observed through a scalar displacement sensor. A C++17 streaming finite impulse response (FIR) core is paired with Python experiments and independent numerical references. At a sampling-rate reduction from 2048 to 256 Hz, a 185 Hz component aliases onto a 71 Hz component. The unfiltered zero-phase amplitude is 0.95000 mm instead of 0.35 mm. A 257-tap Blackman-windowed sinc yields 0.01221% relative amplitude error with 62.5 ms group delay. A phase sweep and explicit error budget show why stopband rejection alone is insufficient. The results verify discrete signal processing on synthetic data; they do not establish experimental physical validity.

1 Research question and governing model

The question is whether the amplitude of a desired mode remains trustworthy when an unobserved higher mode folds onto its sampled frequency. Uniform sampling identifies frequencies modulo the sampling rate; information already lost at acquisition cannot be recovered by subsequent digital filtering [1]. AliasGuard studies a second sampling operation: an initially resolved signal is filtered and decimated for storage or analysis. The work implements established sampling and FIR principles, rather than proposing a new filter family.

Consider three independent, unit-modal-mass oscillators, with amplitudes already projected onto the displacement sensor:

$$\ddot{q}_j + \omega_j^2 q_j = 0, \quad q_j(0) = A_j \cos\phi_j, \quad \dot{q}_j(0) = -A_j\omega_j \sin\phi_j, \qquad x(t) = \sum_{j=1}^{3} q_j(t). \tag{1}$$

Thus $q_j(t) = A_j \cos(\omega_j t + \phi_j)$ exactly. The frequencies $\omega_j/(2\pi) = (23, 71, 185)$ Hz and amplitudes $(1, 0.35, 0.6)$ mm define a synthetic finite-dimensional mechanical model. There is no damping, forcing, measurement noise or spatial discretization. These parameters illustrate a sampling ambiguity and are not calibrated to a particular structure.

2 FIR method and alias budget

For input rate $f_s = 2048$ Hz and integer reduction $R = 8$, the output rate is $F = f_s/R = 256$ Hz. With odd tap count L, delay $D = (L-1)/2$ and cutoff $f_c = 96$ Hz, define the normalized windowed-sinc coefficients [2]

$$\begin{aligned} w_k &= 0.42 - 0.5\cos\frac{2\pi k}{L-1} + 0.08\cos\frac{4\pi k}{L-1}, \\ \widetilde{h}_k &= \frac{2f_c}{f_s}\,\mathrm{sinc}\left(\frac{2f_c}{f_s}(k-D)\right) w_k, \quad h_k = \frac{\widetilde{h}_k}{\sum_\ell \widetilde{h}_\ell}, \quad y_m = \sum_{k=0}^{L-1} h_k x_{mR-k}. \end{aligned} \tag{2}$$

Here $\mathrm{sinc}(z) = \sin(\pi z)/(\pi z)$. The cutoff is approximately a half-amplitude location, not a specified passband edge. Unit DC gain does not imply unit gain at 71 Hz. A longer filter narrows the transition but increases delay and retained-output work.

The causal response is $H(f) = \sum_k h_k \exp(-2\pi i f k/f_s)$. Symmetry makes $g(f) = \exp(2\pi i f D/f_s)H(f)$ real up to roundoff. Output times are relabeled by their filter centers; this removes phase delay in the analysis, not physical latency. On the centered output lattice, 185 Hz folds to $256 - 185 = 71$ Hz with phase reversal. With $\phi_1 = \phi_2 = 0$ and $\phi_3 = \phi$, the colliding phasor is

$$B(\phi) = 0.35g(71) + 0.6g(185)e^{-i\phi}, \qquad \big|\,|B(\phi)| - 0.35\,\big| \le \underbrace{0.35|g(71) - 1|}_{\text{passband distortion}} + \underbrace{0.6|g(185)|}_{\text{residual alias}}. \tag{3}$$

The bound follows from the reverse triangle inequality and is valid for all phases. Without filtering, $|B|$ ranges from $|0.35 - 0.6| = 0.25$ to 0.95 mm. Those data alone cannot identify the two original modal amplitudes.

3 Implementation and reproducible experiment

The C++ core maintains a circular history buffer and a reduction phase. Each input updates the buffer; only phase-zero samples trigger the dot product in Eq. (2). Phase persists across calls, including irregular chunks. The cost is $O(N + NL/R)$ operations for N inputs and $O(L)$ state. The pybind11 interface copies finite, real arrays. There is no FFT, performance benchmark or real-time guarantee. Invalid numerical chunks are rejected before state changes; empty chunks are no-ops, and reset clears both history and phase. Zero prehistory is the core default, with no automatic trimming or tail flushing.

The experiment evaluates Eq. (1) over the true prehistory needed by each filter, discards startup and retains 2048 centered outputs at $t_m = m/256$ s, $m = 0, \ldots, 2047$. This is an 8 s record with a last sample before 8 s. Tap counts are 65, 129, 257 and 513. All delays are integer multiples of the output interval, so no fractional resampling is hidden in the alignment. The signal is fed in chunks split after 37 and 1082 inputs, including an empty call. For each L, the measured 71 Hz amplitude comes from least-squares fitting of a DC term plus sine/cosine pairs at 23 and 71 Hz. Frequencies are known; this is not a frequency-discovery algorithm. A second experiment uses 65 equally spaced phases over $[0, 2\pi]$, including the duplicate periodic endpoint, at 257 taps.

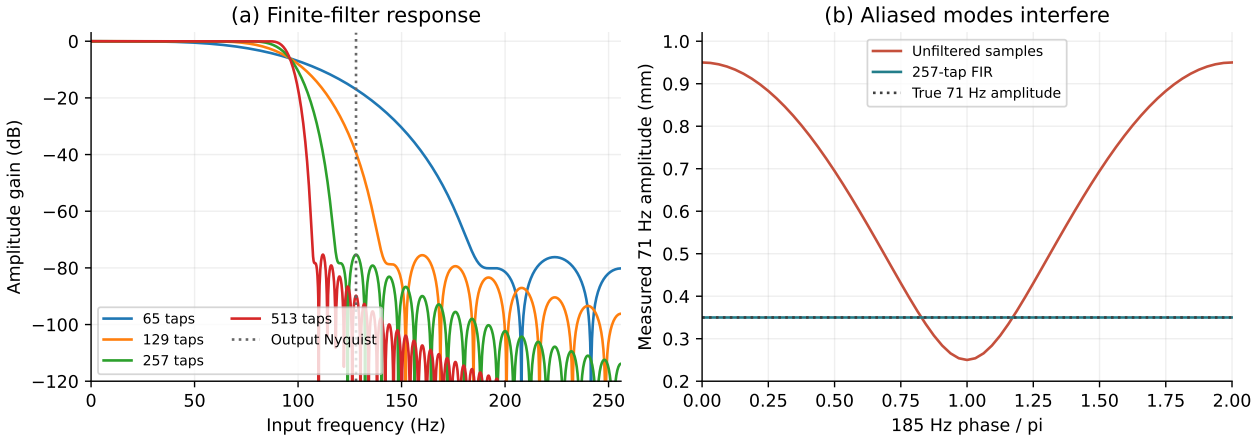


Figure 1: Project-generated frequency responses and measured modal interference. The finite-filter transition affects the desired 71 Hz mode. The phase sweep reveals an amplitude ambiguity after unfiltered sampling; the 257-tap output stays close to the prescribed desired amplitude.

Run `aliasguard -output examples/reproduced` after installing the package. The committed JSON and three CSV files contain the setup, all filter and phase results, and the full time record. The paper's numerical macros are generated from that JSON; input and PDF hashes bind the manuscript to the source and data.

4 Measured results and independent verification

Table 1 reports the zero-phase case. Low-band RMSE compares with $x_{\mathrm{low}}(t) = \cos(2\pi 23t) + 0.35\cos(2\pi 71t)$ mm, not with the full three-mode signal. Without filtering, this RMSE is 0.42426 mm. A 65-tap filter nearly removes the high mode but loses substantial desired-mode amplitude. At 257 taps, the residual-alias bound is 7.015×10^{-6} mm while the passband bound is 4.975×10^{-5} mm; the latter is larger. Over all sampled phases, the measured amplitude lies between 0.34994323 and 0.34995726 mm.

Table 1: Measured filter-length study. Signed $g(185)$ is retained because its sign changes the interference. RMSE is relative to the prescribed low-band signal.

L	Delay (ms)	$g(71)$	$g(185)$	Amplitude (mm)	RMSE (mm)
65	15.625	0.7972680	1.640×10^{-4}	0.2791422	5.015×10^{-2}
129	31.250	0.9577400	-4.246×10^{-6}	0.3352065	1.046×10^{-2}
257	62.500	0.9998579	1.169×10^{-5}	0.3499573	4.474×10^{-5}
513	125.000	1.0000485	-1.169×10^{-6}	0.3500163	1.152×10^{-5}

Independent SciPy implementations [3] verify causal convolution, Blackman-windowed design and frequency response. Across the four filters, maximum differences are 3.331×10^{-15} mm, 1.388×10^{-17} in coefficients, and 3.986×10^{-15} in complex response. An analytical harmonic prediction using the centered transfer gains agrees with the retained time samples within 2.714×10^{-13} mm. The independent phasor expression in Eq. (3) agrees with measured amplitudes over the phase sweep within 1.457×10^{-13} mm. These local values are recorded outputs, not universal roundoff bounds.

Tests additionally verify phase reversal, startup, reset, chunk invariance, DC preservation, malformed inputs and the amplitude bound. Reproduction permits relative tolerance 2×10^{-7} and absolute tolerance 5×10^{-12} in each stored field's units across compilers/libraries. CI tests native and Python builds and installed wheels on Windows and Ubuntu; Linux also checks address and undefined-behavior sanitizers. Its independent pdfLaTeX manuscript build enforces at most five pages, resolves citations and rejects overfull boxes. Rendered pages are inspected separately from automated numerical checks.

5 Discussion, limitations and conclusion

Three distinctions matter. First, differences against convolution and phasor references measure *discrete implementation error*. Second, low-band RMSE and amplitude bias measure *processing distortion*; even an exactly implemented FIR deliberately changes the input. Third, no *continuum error* or physical-validation claim follows: there is no PDE mesh and the prescribed modal trajectories are evaluated analytically. An actual structure could contain unmodeled modes, damping and nonlinearities. Sensor noise, ADC quantization, uncertain frequencies and prior acquisition aliasing are omitted.

Increasing length does not monotonically decrease gain at a single stopband frequency: 129 taps happen to reject 185 Hz more strongly than 257 taps, while preserving 71 Hz less accurately. Choosing a filter therefore requires passband and stopband criteria together with acceptable latency. A finite FIR cannot guarantee rejection of every signal beyond output Nyquist. The synthetic coherent record and known-frequency regression do not assess blind modal identification.

AliasGuard's main result is a reproducible two-part amplitude-error account. For this model, filtering before decimation largely preserves the desired mode, whereas downsampling first produces an irreducible ambiguity. The analytical bound makes the tradeoff between wanted-mode distortion and residual aliasing explicit; it must be reassessed for other spectra and acquisition chains.

References

[1] S. W. Smith. *The Scientist and Engineer's Guide to Digital Signal Processing*, Ch. 3, "The Sampling Theorem." https://dspguide.com/ch3/2.htm.

[2] S. W. Smith. Same work, Ch. 16, "Designing the Filter." https://www.dspguide.com/ch16/2.htm.

[3] SciPy developers. Signal processing API: `firwin`, `lfilter`, `freqz`. https://docs.scipy.org/doc/scipy/reference/signal.html.